

NexusIQ ■ — Master Technical Architecture

Special Briefing & Presentation Guide for Senior Engineering Advisor: Sumit Sharan (Mama Ji)

Event: InnovaHack Chapter 1 National Hackathon 2026 (Round 2 Finals — Domain 3: Gen AI)
Live App: <https://nexusiq2.vercel.app> | **Backend API:** <https://nexusiq-backend-production.up.railway.app>

1. Executive Problem Statement & Core Value Proposition

In modern enterprises, compliance audits (ISO 27001, GDPR, HIPAA, SOC 2, EU AI Act) require sifting through thousands of multi-modal documents (PDFs, audio recordings, video transcripts, and physical server room hardware telemetry).

The Flaw of Standard Vector RAG: Traditional Enterprise AI uses vector embeddings (chunking text into vector databases). In strict compliance environments, vector RAG fails catastrophically due to:

- **Blind Chunking:** Severing sentences destroys multi-page policy relationships (e.g., *'Section A applies unless overridden by Section B on page 40'*).
- **AI Hallucinations:** Cosine similarity pulls 'sounds related' chunks, leading the LLM to invent non-existent rules. In compliance, an AI hallucination leads to immediate legal lawsuits.
- **Physical Disconnect:** Software vector databases are completely blind to real-world physical events (e.g., server room thermal spikes or hardware failures).

The NexusIQ Solution: NexusIQ replaces raw vector lookups with a **Directed Knowledge Graph RAG Engine** paired with a physical **IoT Edge Hardware Feed**. Every policy, person, and hardware device becomes an explicit Node, and every rule becomes a Directed Edge. Queries execute via **2-Hop BFS Subgraph Traversal**, forcing the LLM to provide 100% citation-grounded answers with zero hallucination.

2. System Architecture & Component Breakdown

NexusIQ is built across 5 decoupled production layers designed for zero-latency execution, multi-model resilience, and real-time physical edge telemetry.

Layer	Technology Stack	Role & Function
Layer 1: IoT Edge	ESP32 DevKit V1 + RPi Gateway	Reads DHT11 temp/humidity every 2s. Trips 4-relay fan control at >=32°C. Posts phys
Layer 2: Frontend	React 18, Vite, D3.js, Three.js	Renders 2D force-directed knowledge graph (D3.js), 3D starfield canvas (Three.js), and
Layer 3: Backend	FastAPI, NetworkX, PyMuPDF, SlowAP	Async Python REST API maintaining NetworkX DiGraph in memory. Computes dynam
Layer 4: AI Layer	Lyzr SuperFlow, Gemini 2.5, OpenRouter	Primary: Lyzr AI SuperFlow Agentic Webhook. Multimodal Base: Gemini 2.5. Fallback:
Layer 5: Database	Supabase PostgreSQL, Firebase Auth	Stores full graph state JSON in `nexusiq_graphs` table. Handles Google OAuth & user

3. Detailed System Workflow Architecture

A. Document Ingestion & Graph Extraction Workflow

1. User uploads PDF/Audio/Text via React UI (UploadPanel).
2. FastAPI passes text/file to PyMuPDF or Gemini 1.5 Pro audio transcription engine.

- 3. Google Gemini 2.5 Flash extracts entities & relationships matching explicit JSON schema.
- 4. NetworkX Engine ingests nodes/edges with canonical deduplication & updates Supabase PostgreSQL.
- 5. Server-Sent Events (SSE) stream `upload` signal to frontend -> D3.js graph animates live.

B. Zero-Hallucination Graph RAG Query Workflow

- 1. User types natural language question in chat interface.
- 2. Backend extracts target entity keywords & executes 2-Hop Breadth-First Search (BFS) in NetworkX graph.
- 3. Subgraph context is formatted into grounded assertion block: *[SOURCE: ISO_27001.pdf] Node A -> GOVERNS -> Node B.*
- 4. Grounded block is sent to Lyzr AI SuperFlow webhook for deterministic answer generation.
- 5. Answer is parsed for citations. Hallucination metric updates (0.0% rate maintained).

C. Physical IoT Edge Hardware Incident Workflow

- 1. ESP32 DHT11 sensor monitors server room temperature every 2 seconds.
- 2. If temp >= 32°C, ESP32 trips 4-channel active-low relay (emergency fan shutdown) and POSTs alert to RPi on port 5001.
- 3. Raspberry Pi forwards payload to Cloud Backend (`POST /upload-text`).
- 4. Backend creates red Alert Node, penalizes Compliance Readiness score (-10%), and pushes SSE event to live feed.

4. Mathematical Models & Scoring Formulas

Dynamic Compliance Readiness Score Formula:

Score = Clamp(50 + (5 × F) + (2 × P) + (1 × R) - (10 × C) - (5 × V) , 0 , 100)

- **F** = Count of recognized Compliance Framework entities (ISO 27001, GDPR, HIPAA, etc.)
- **P** = Count of Policy nodes in graph | **R** = Count of verified Relationship edges
- **C** = Count of Critical Thermal/Security Incident nodes (-10% penalty per incident)
- **V** = Count of unresolved Violation nodes (-5% penalty per violation)

5. 5-Minute Pitch Script & Strategy (For Sumit Sharan)

As Senior Engineering Advisor & Lead Presenter, use this step-by-step 5-minute demonstration script to captivate the judges:

Time	Action on Screen	Exact Script to Memorize & Deliver
0:00 - 1:00	Show Landing Page (nexusiq.vercel.app)	Honorable judges, traditional enterprise AI fails in compliance because vector chunking destroys
1:00 - 2:00	Click 'Enter Workspace' & point to StatsBar	Notice our top metrics bar. Our Compliance Score (87.5%) and Hallucination Rate (0.0%) are no
2:00 - 3:00	Drag & drop ISO 27001 PDF into Upload Panel	Watch what happens when we upload an ISO 27001 document. Google Gemini extracts entities
3:00 - 4:00	Type in chat: 'What are our retention policies?'	When we ask a question, NexusIQ runs a 2-hop BFS on our graph and passes context to Lyzr AI
4:00 - 5:00	Click 'Trigger Edge Incident' & Export Report	Compliance is not just digital; it's physical. We connected custom ESP32 hardware. When serve

Team NexusIQ: Hemant Prakash (Lead Full Stack & AI Architect) | Shubham Kumar (QA Lead) | Sumit Sharan / Mama Ji (Senior Engineering Advisor, Developer & Presentation Lead)